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import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import joblib

# Load data

df = pd.read_csv("data.csv")

# Duplicate check

n_duplicates = df.duplicated().sum()
print(f"Duplicate rows found: {n_duplicates}")
if n_duplicates > 0:

df = df.drop_duplicates()
print(f"Duplicates dropped. Rows remaining: {len(df)}")
```

```
# Handling missing values
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```
missing_before = df.isna().sum()
total_missing = missing_before.sum()
rows_before = len(df)

print("Missing values by column:")
print(missing_before[missing_before > 0])
print(f"\nTotal missing cells: {total_missing}")
```

```
# Check whether missingness is concentrated in a particular region or
# risk group before blindly dropping it – a silent drop can bias the
# training population if missingness isn't random.
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```
if total_missing > 0 and "region" in df.columns:
    rows_with_na = df[df.isna().any(axis=1)]
    print("\nMissingness by region (rows with at least one NaN):")
    print(rows_with_na["region"].value_counts())
```

```
df = df.dropna()
rows_after = len(df)
pct_dropped = 100 * (rows_before - rows_after) / rows_before
print(f"\nRows before drop: {rows_before} | after: {rows_after} "
      f"| dropped: {rows_before - rows_after} ({pct_dropped:.2f}%)")
```

```
if pct_dropped > 5:
    print("WARNING: more than 5% of rows dropped for missingness – "
          "confirm this isn't concentrated in one segment before proceeding.")
```

```
assert df.isna().sum().sum() == 0, "Missing values still present after dropna()"
```

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# Summary statistics
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print(df.describe(include="all"))
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